2020-2021 (SPRING) FACULTY of CIV. ENG. – DIV. of MECHANICS, STRENGTH OF MATERIALS II

FINAL EXAM *17.06.2021*

Name:
Group:
Sign:
No:

Explanation:

Number a: is the last digit of your student number.
Number b: is the second last digit of your student number.

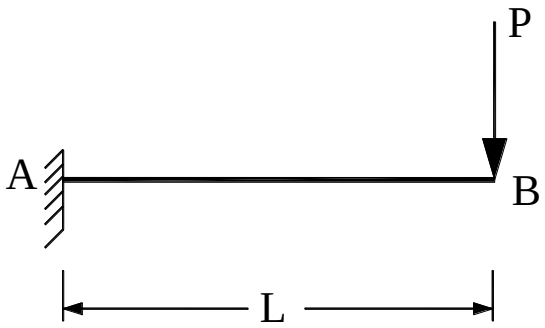
Example for **"L=1ab (cm)":** Student number: xxxxxx94, a=4, b=9 then **L=149 cm**

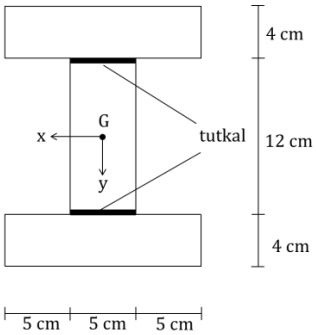
Duration: **29 minutes** (solution + uploading)

1) (25 p)

A wooden beam having a span length of L is built with three timber parts connecting by glue and supported as shown in the figure. Knowing that the allowable stresses for the material of the beam are $\tau_{all} = 0.8\ MPa$, $\sigma_{all} = 14\ MPa$ and safety shear stress for the glue is $\tau_{all,glue} = 0.4\ MPa$,

a- find the load **P** that can be safely carried by the beam,
b- using the **P** value you obtained, draw the normal and shear stress diagrams in the most critical section of the beam.





(cross-section)

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Number c: is the third last digit of your student number.

Example for **"M_o=a+b+c (Ncm)":** Student number: xxxxxx94, *a*=4, *b*=9, c=3 then **M_o=4+9+3=16 Ncm**

Example for **"L=1ab (cm)":** Student number: xxxxxx94, *a*=4, *b*=9 then **L=149 cm**

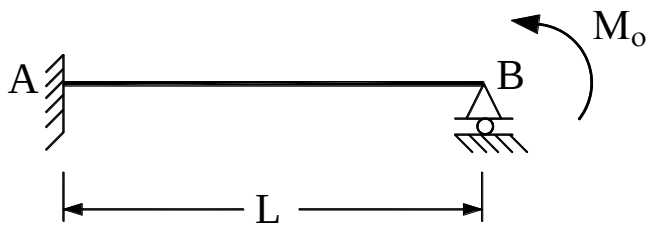
Duration: **29 minutes** (solution + uploading)

2) (25 p)

Find the elastic curve equation of the statically indeterminate beam given in the figure.

Draw $\left(T_y\right)$ and $\left(M_x\right)$ diagrams of the beam.

The bending rigidity EI for the beam is constant along the axis of the beam.



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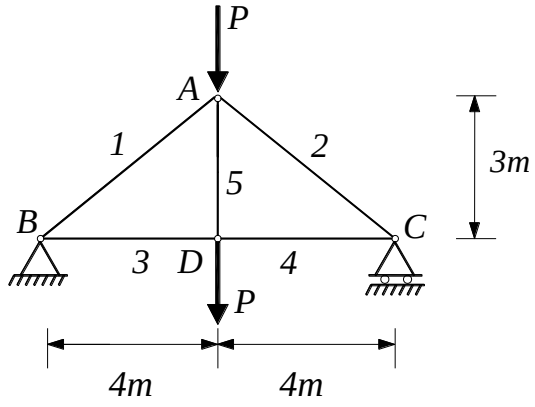
Number a: is the last digit of your student number.
Number b: is the second last digit of your student number.
Number c: is the third last digit of your student number.

- Example for **"P=a+b+c (N)":** Student number: xxxxxx94, *a*=4, *b*=9, c=3 then **P=4+9+3=16 N**

Duration: **29 minutes** (solution + uploading)

3) (25 p)

The elongation stiffness of the bars is in the truss system shown in the figure is constant and equal to EA. Find the vertical displacement at point **D** by using the CASTIGLIANO theorem.



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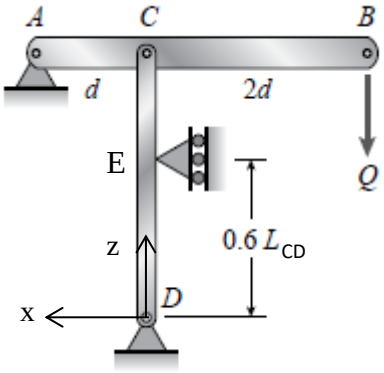
Explanation:

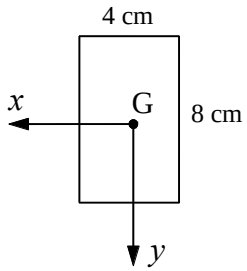
Q (kN) : Summation of digits in the student ID number

Duration: **29 minutes** (solution + uploading)

4) (25 p)

For the frame system given in the figure, the movement of the point E in the (x-z) plane is restrained, but its movement in the (y-z) plane is allowed. Determine if the frame can support the given loading according to the buckling. $L_{CD} = 2m$, $d = L_{CD}/2$, $n = 2$, $E = 210\ GPa$, $\lambda_p = 105$, $\sigma_{cr} = 310 - 1.14\lambda\ (MPa)$





(Cross-section of the column CD)